

HEADERS

IntroCppGamedev *.h - Everything	
File Edit View Search Index History Bookmarks Tools Help	
IntroCppGamedev *.h	
Name	Path
dino_animal.h	C:\Users\j.vernay\Desktop\IntroCppGamedev\src\dino
dino_draw_utils.h	C:\Users\j.vernay\Desktop\IntroCppGamedev\src\dino
dino_geometry.h	C:\Users\j.vernay\Desktop\IntroCppGamedev\src\dino
dino_main.h	C:\Users\j.vernay\Desktop\IntroCppGamedev\src\dino
dino_scene.h	C:\Users\j.vernay\Desktop\IntroCppGamedev\src\dino
dino_terrain.h	C:\Users\j.vernay\Desktop\IntroCppGamedev\src\dino
exo2_draw.h	C:\Users\j.vernay\Desktop\IntroCppGamedev\src\exo2
exo2_pong.h	C:\Users\j.vernay\Desktop\IntroCppGamedev\src\exo2
exo4_image.h	C:\Users\j.vernay\Desktop\IntroCppGamedev\src\exo4
exo5_stats.h	C:\Users\j.vernay\Desktop\IntroCppGamedev\src\exo5
jv.h	C:\Users\j.vernay\Desktop\IntroCppGamedev\src\jv
jv_win64_impl.h	C:\Users\j.vernay\Desktop\IntroCppGamedev\src\jv\x64-windows

SOURCES

IntroCppGamedev *.cpp - Everything	
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IntroCppGamedev *.cpp	
Name	Path
dino_animal.cpp	C:\Users\j.vernay\Desktop\IntroCppGamedev\src\dino
dino_draw_utils.cpp	C:\Users\j.vernay\Desktop\IntroCppGamedev\src\dino
dino_geometry.cpp	C:\Users\j.vernay\Desktop\IntroCppGamedev\src\dino
dino_main.cpp	C:\Users\j.vernay\Desktop\IntroCppGamedev\src\dino
dino_scene.cpp	C:\Users\j.vernay\Desktop\IntroCppGamedev\src\dino
dino_terrain.cpp	C:\Users\j.vernay\Desktop\IntroCppGamedev\src\dino
exo1_pong.cpp	C:\Users\j.vernay\Desktop\IntroCppGamedev\src\exo1
exo2_draw.cpp	C:\Users\j.vernay\Desktop\IntroCppGamedev\src\exo2
exo2_main.cpp	C:\Users\j.vernay\Desktop\IntroCppGamedev\src\exo2
exo2_pong.cpp	C:\Users\j.vernay\Desktop\IntroCppGamedev\src\exo2
exo4_image.cpp	C:\Users\j.vernay\Desktop\IntroCppGamedev\src\exo4
exo4_main.cpp	C:\Users\j.vernay\Desktop\IntroCppGamedev\src\exo4
exo5_main.cpp	C:\Users\j.vernay\Desktop\IntroCppGamedev\src\exo5
exo5_stats.cpp	C:\Users\j.vernay\Desktop\IntroCppGamedev\src\exo5
exo6_main.cpp	C:\Users\j.vernay\Desktop\IntroCppGamedev\src\exo6
jv_random.cpp	C:\Users\j.vernay\Desktop\IntroCppGamedev\src\jv\common
jv_win64_file.cpp	C:\Users\j.vernay\Desktop\IntroCppGamedev\src\jv\x64-windows
jv_win64_main.cpp	C:\Users\j.vernay\Desktop\IntroCppGamedev\src\jv\x64-windows
jv_win64_renderer.cpp	C:\Users\j.vernay\Desktop\IntroCppGamedev\src\jv\x64-windows
jv_win64_shaders.cpp	C:\Users\j.vernay\Desktop\IntroCppGamedev\src\jv\x64-windows
jv_win64_window.cpp	C:\Users\j.vernay\Desktop\IntroCppGamedev\src\jv\x64-windows

src ✓

OBJECTS

IntroCppGamedev *.obj - Everything

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IntroCppGamedev *.obj

Name	Path
exo1_pong.obj	C:\Users\j.vernay\Desktop\IntroCppGamedev\build\obj\x64-windows\Debug\Exo1
exo2_draw.obj	C:\Users\j.vernay\Desktop\IntroCppGamedev\build\obj\x64-windows\Debug\Exo2
exo2_main.obj	C:\Users\j.vernay\Desktop\IntroCppGamedev\build\obj\x64-windows\Debug\Exo2
exo2_pong.obj	C:\Users\j.vernay\Desktop\IntroCppGamedev\build\obj\x64-windows\Debug\Exo2
exo5_main.obj	C:\Users\j.vernay\Desktop\IntroCppGamedev\build\obj\x64-windows\Release\Exo5
exo5_stats.obj	C:\Users\j.vernay\Desktop\IntroCppGamedev\build\obj\x64-windows\Release\Exo5
jv_random.obj	C:\Users\j.vernay\Desktop\IntroCppGamedev\build\obj\x64-windows\Debug\JV
jv_win64_file.obj	C:\Users\j.vernay\Desktop\IntroCppGamedev\build\obj\x64-windows\Debug\JV
jv_win64_main.obj	C:\Users\j.vernay\Desktop\IntroCppGamedev\build\obj\x64-windows\Debug\JV
jv_win64_renderer.obj	C:\Users\j.vernay\Desktop\IntroCppGamedev\build\obj\x64-windows\Debug\JV
jv_win64_shaders.obj	C:\Users\j.vernay\Desktop\IntroCppGamedev\build\obj\x64-windows\Debug\JV
jv_win64_window.obj	C:\Users\j.vernay\Desktop\IntroCppGamedev\build\obj\x64-windows\Debug\JV

→ Nom de fichiers C++

LIB

IntroCppGamedev *.lib

Name	Path
JV.lib	C:\Users\j.vernay\Desktop\IntroCppGamedev\build\x64-windows\Debug

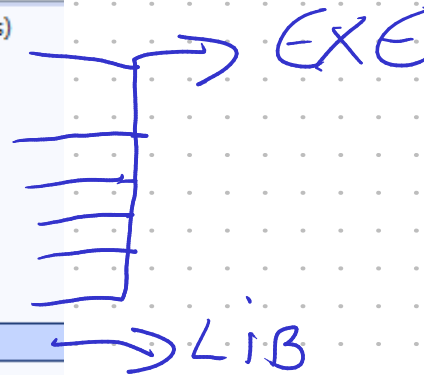
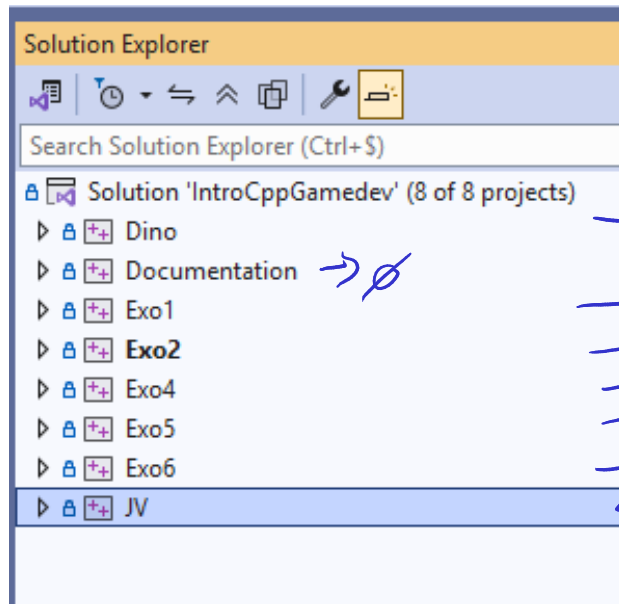
EXE

IntroCppGamedev *.exe !tools - Everything

File Edit View Search Index History Bookmarks Tools Help

IntroCppGamedev *.exe !tools

Name	Path
Dino.exe	C:\Users\j.vernay\Desktop\IntroCppGamedev
Exo1.exe	C:\Users\j.vernay\Desktop\IntroCppGamedev\build\x64-windows\Debug
Exo2.exe	C:\Users\j.vernay\Desktop\IntroCppGamedev\build\x64-windows\Debug



```
PS C:\Users\j.vernay\Desktop\IntroCppGamedev\build> lib /list .\x64-windows\Debug\JV.lib | findstr obj
C:\Users\j.vernay\Desktop\IntroCppGamedev\build\obj\x64-windows\Debug\JV\jv_random.obj
C:\Users\j.vernay\Desktop\IntroCppGamedev\build\obj\x64-windows\Debug\JV\jv_win64_file.obj
C:\Users\j.vernay\Desktop\IntroCppGamedev\build\obj\x64-windows\Debug\JV\jv_win64_main.obj
C:\Users\j.vernay\Desktop\IntroCppGamedev\build\obj\x64-windows\Debug\JV\jv_win64_renderer.obj
C:\Users\j.vernay\Desktop\IntroCppGamedev\build\obj\x64-windows\Debug\JV\jv_win64_shaders.obj
C:\Users\j.vernay\Desktop\IntroCppGamedev\build\obj\x64-windows\Debug\JV\jv_win64_window.obj
d:\os\obj\amd64fre\onecoreuap\windows\directx\misc\dxguid\daytona\objfre\amd64\d3d10guid.obj
d:\os\obj\amd64fre\onecoreuap\windows\directx\misc\dxguid\daytona\objfre\amd64\d3d9guid.obj
d:\os\obj\amd64fre\onecoreuap\windows\directx\misc\dxguid\daytona\objfre\amd64\dxguid.obj
```

LIB
contient
des OBJs

À Chaque projet correspond un fichier VCXPROJ dans le dossier build qui déclare à Visual Studio comment le projet est compilé

```
<ItemGroup>
  <ClInclude Include="..\src\exo2\exo2_draw.h" />
  <ClInclude Include="..\src\exo2\exo2_pong.h" />
</ItemGroup>
<ItemGroup>
  <ClCompile Include="..\src\exo2\exo2_draw.cpp" />
  <ClCompile Include="..\src\exo2\exo2_main.cpp" />
  <ClCompile Include="..\src\exo2\exo2_pong.cpp" />
</ItemGroup>
```

→ Sont transformés en OBJ

```
<PropertyGroup Condition="'$(Configuration)|$(Platform)'=='Debug x64-windows|x64'">
  <LinkIncremental>>false</LinkIncremental>
  <OutDir>$(ProjectDir)x64-windows\Debug</OutDir>
  <IntDir>$(ProjectDir)obj\x64-windows\Debug\Exo2</IntDir>
  <TargetName>Exo2</TargetName>
  <TargetExt>.exe</TargetExt>
</PropertyGroup>
```

→ Destination du fichier EXE

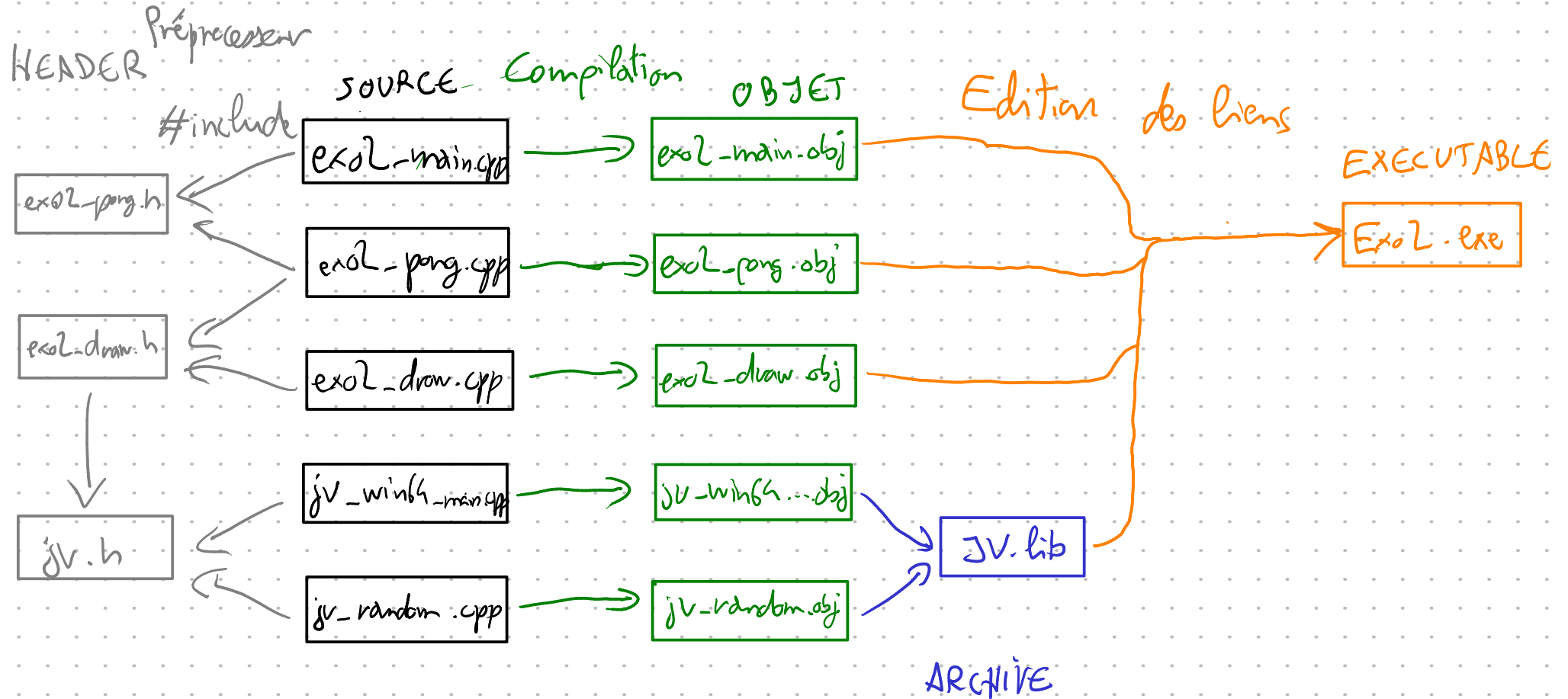
→ Destination des fichiers OBJ

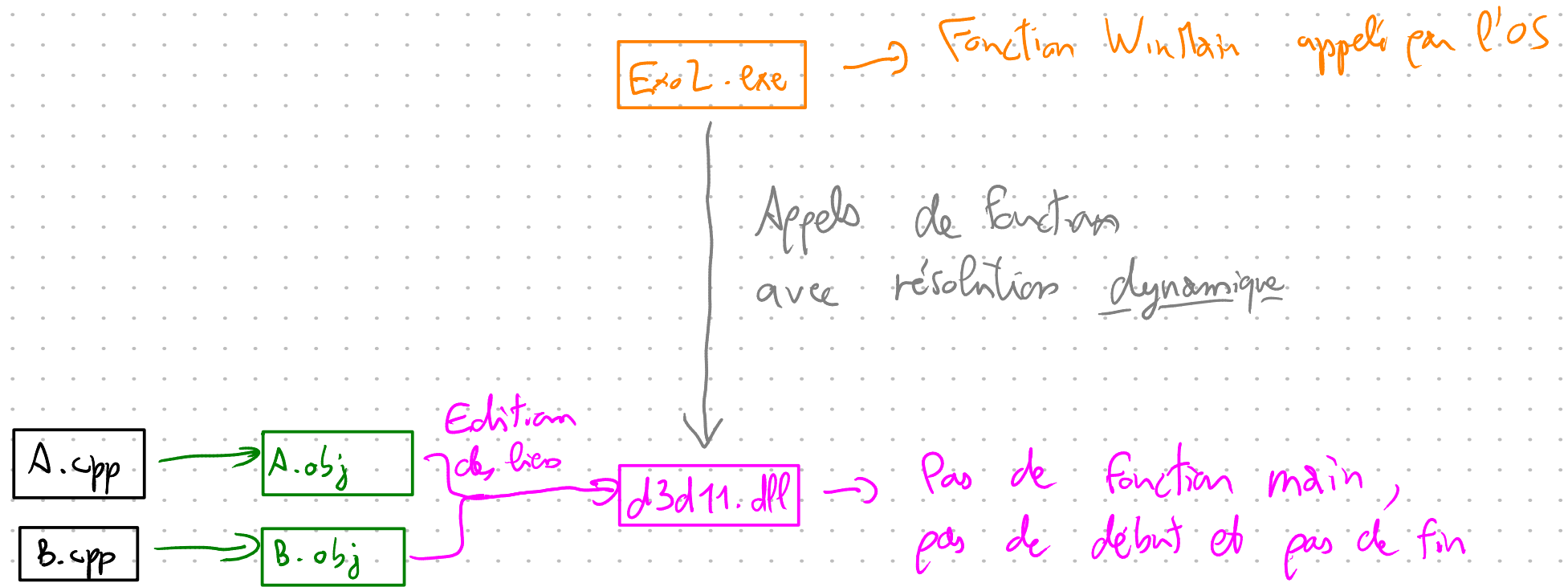
```
<LanguageStandard>stdcpp20</LanguageStandard>
```

```
<AdditionalIncludeDirectories>..\src;%(AdditionalIncludeDirectories)</AdditionalIncludeDirectories>
```

→ où le compilateur cherche les fichiers à inclure

.H , .CPP , .OBJ , .LIB , .EXE





⇒ La résolution de liens avec une DLL est faite à l'exécution.
LIB → à la compilation

- LIB = Archive = Librairie statique
- DLL = Librairie dynamique

Deux intérêts de la DLL :

1. On peut mettre à jour les fichiers EXE et DLL indépendamment
2. On peut partager du code entre plusieurs EXE
→ Permet d'économiser de l'espace disque

PRÉPROCESSEUR

A.h

```
int  mafunc (
```

B.h

```
return 5
```

Main.cpp

```
#include <A.h>
} {
#include <B.h>
; }

int v = mafunc();
```

Une directive de préprocesseur commence par un # :

#include

#define

#if

etc

```
// Destination
```

```
std::vector<jv::util::Color> pixels;
```

```
pixels.resize(1);
```

```
jv::util::Color* pPixels;
```

```
// A ENLEVER
```

```
float f = jv::util::Rand
```

```
-Divide[50] = 128;
```

 union jv::util::Color

Représente une couleur de pixels, utilisée pour moduler l'affichage d'une texture.

Size: 4 bytes

Alignment: 4 bytes

[Search Online](#) | [Memory Layout](#)

Taille du type

```
/// Représente une couleur de pixels, utilisée pour moduler l'affichage d'une texture.
```

```
union Color {
```

```
    struct
```

```
    {
```

```
        uint8_t r;
```

```
        uint8_t g;
```

```
        uint8_t b;
```

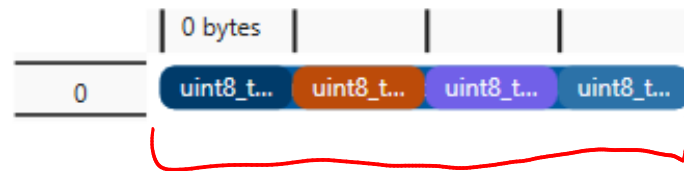
```
        uint8_t a;
```

```
    };
```

```
    uint32_t rgba;
```

```
};
```

union jv::util::Color



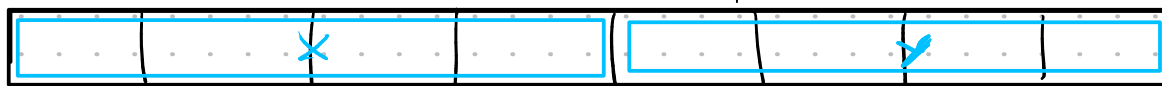
*4 x uint8_t
= 4 octets*

union jv::util::Color

Offset	Field name	Size
0	uint32_t jv::util::Color::rgba	4 bytes
0	uint8_t jv::util::Color::r	1 bytes
1	uint8_t jv::util::Color::g	1 bytes
2	uint8_t jv::util::Color::b	1 bytes
3	uint8_t jv::util::Color::a	1 bytes

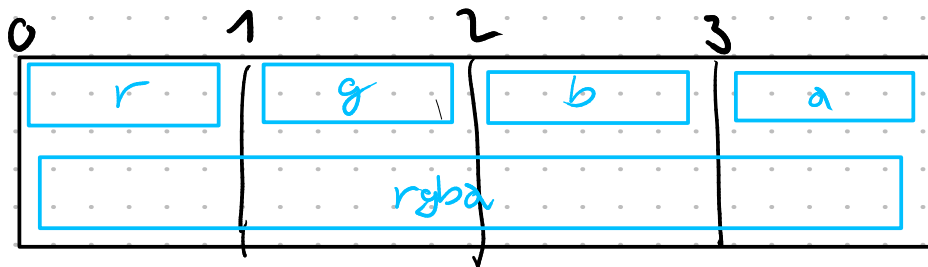
8 octets ?

struct Vec2 { float x, y; };



Taille de 8 octets

Union Color



Total 4 offset

union jv::util::Color



Offset	Field name	Size
0	uint32_t jv::util::Color::rgba	4 bytes
0	uint8_t jv::util::Color::r	1 bytes
1	uint8_t jv::util::Color::g	1 bytes
2	uint8_t jv::util::Color::b	1 bytes
3	uint8_t jv::util::Color::a	1 bytes

En C et C++,
char

≠ caractère au sens commun

char = un octet = la plus petite unité adressable

un octet $\rightarrow$ 8 bits $\rightarrow 2^8$ valeurs $\rightarrow$ 256 valeurs

$[-128; 127]$ $\rightarrow$ signed char

$[0; 255]$ $\rightarrow$ unsigned char

Type standard

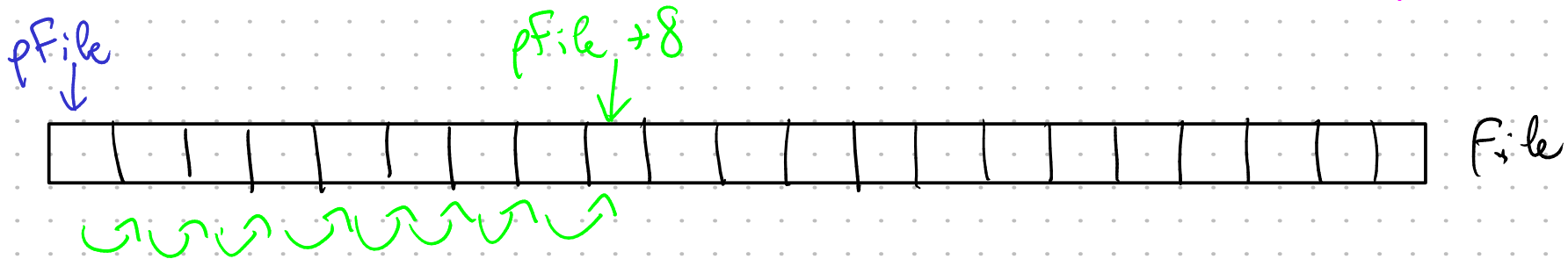
std::vector <unsigned char>

$\rightarrow$ un tableau d'octets

$\hookrightarrow$ Tableau / Array dynamique

```
unsigned char* pFile = file.data();
```

Pointeur vers un octet
Pointeur vers ...
méthode data() de std::vector
Pointeur vers la première case du tableau
(ici, le premier octet)



```
*(uint32_t*)(pFile + 0x0012)
```

Pointeur vers premier octet
Pointeur vers le 18^{ème} octet
Pointeur vers un uint32 à l'emplacement du 18^{ème} octet
Lire un uint32 à l'emplacement du 18^{ème} octet

Watch 1		
Search (Ctrl+E)    Search Depth: 3  		
Name	Value	Type
▸  pFile	0x000002a4d7a37080 "BMN-\x15"	 View ▾ unsigned char *
▸  pFile + 1	0x000002a4d7a37081 "MN-\x15"	 View ▾ unsigned char *
▸  pPixels	0x000002a4d754e720 {r=128 '€' g=82 'R' b=8 '\b' ...}	<u>jav::util::Color *</u>
▸  pPixels + 1	0x000002a4d754e724 {r=253 'ý' g=253 'ý' b=253 'ý' ...}	<u>jav::util::Color *</u>
 sizeof(unsigned char)	1	unsigned __int64
 sizeof(jav::util::Color)	4	unsigned __int64

`pPixels[0].r = 128;`

↳ Idx de case
↳ Pointeur vers une couleur

« Je veux accéder à la 0-ième case à partir de pPixels »

$\Leftrightarrow *(\text{pPixels} + N)$

$\Leftrightarrow *(N + \text{pPixels})$

$\Leftrightarrow N[\text{pPixels}]$

Fichier BMP, comme tout fichier, est une suite d'octets
 Un format de fichier dicte comment interpréter les octets -

Name	Size	Offset	Description
Header	14 bytes		Windows Structure: BITMAPFILEHEADER
Signature	2 bytes	0000h	'BM'
FileSize	4 bytes	0002h	File size in bytes
reserved	4 bytes	0006h	unused (=0)
DataOffset	4 bytes	000Ah	Offset from beginning of file to the beginning of the bitmap data
InfoHeader	40 bytes		Windows Structure: BITMAPINFOHEADER
Size	4 bytes	000Eh	Size of InfoHeader = 40
Width	4 bytes	0012h	Horizontal width of bitmap in pixels → v1
Height	4 bytes	0016h	Vertical height of bitmap in pixels → v2
Planes	2 bytes	001Ah	Number of Planes (=1)
Bits Per Pixel	2 bytes	001Ch	Bits per Pixel used to store palette entry information. This also identifies in an indirect way the number of possible colors. Possible values are: 1 = monochrome palette. NumColors = 1 4 = 4bit palletized. NumColors = 16 8 = 8bit palletized. NumColors = 256 16 = 16bit RGB. NumColors = 65536 24 = 24bit RGB. NumColors = 16M → v3
Compression	4 bytes	001Eh	Type of Compression 0 = BI_RGB no compression 1 = BI_RLE8 8bit RLE encoding 2 = BI_RLE4 4bit RLE encoding
ImageSize	4 bytes	0022h	(compressed) Size of Image It is valid to set this =0 if Compression = 0
XpixelsPerM	4 bytes	0026h	horizontal resolution: Pixels/meter
YpixelsPerM	4 bytes	002Ah	vertical resolution: Pixels/meter
Colors Used	4 bytes	002Eh	Number of actually used colors. For a 8-bit / pixel bitmap this will be 100h or 256.
Important Colors	4 bytes	0032h	Number of important colors 0 = all
ColorTable	4 * NumColors bytes	0036h	present only if Info.BitsPerPixel less than 8 colors should be ordered by importance
Red	1 byte		Red intensity
Green	1 byte		Green intensity
Blue	1 byte		Blue intensity
reserved	1 byte		unused (=0)
repeated NumColors times			
Pixel Data	InfoHeader.ImageSize bytes		The image data

BMP

Data Offset

Header, Offset 0, Taille 14

InfoHeader, Offset 14, Taille 40

ça dépend...

Pixel Data, Offset?

Si 24 bits par pixel = 3 octets par pixel

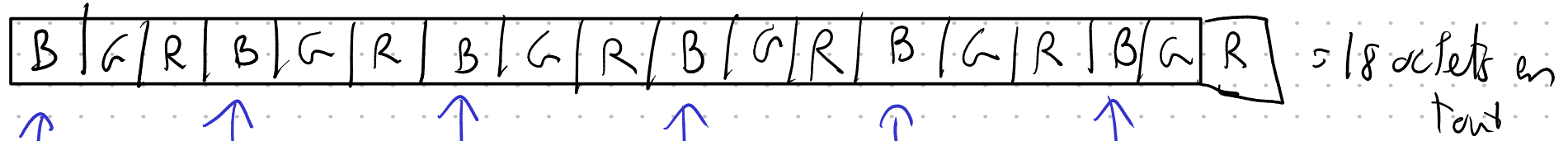
Taille = 3 octet $\times$ largeur $\times$ hauteur
3 $\times$ pxWidth $\times$ pxHeight

B	G	R	B	G	R	B	G	R	B	G	R	B	G	R
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

Pixels ...

pxWidth
pxHeight
bits per pixel

Taille Width = 2, Height = 3 $\rightarrow$ Width $\times$ Height = 6 pixels en tout



Fichier BMP

Width pixels
par ligne

E	F
C	D
A	B

VRAM

[0]	[1]
[2]	[3]
[4]	[5]

VRAM

[0]	E : R G B 255
[1]	F : R G B 255
[2]	C : R G B 255
[3]	D : R G B 255
[4]	A : R G B 255
[5]	B : R G B 255

BMP

Data Offset

Header, Offset 0, Taille 14

InfoHeader, Offset 14, Taille 40

Color Table, Offset 54, Taille ?

B ₀	G ₀	R ₀	...	B ₁	G ₁	R ₁	...	B ₂	G ₂	R ₂	...	B ₃	G ₃	R ₃	...	etc
----------------	----------------	----------------	-----	----------------	----------------	----------------	-----	----------------	----------------	----------------	-----	----------------	----------------	----------------	-----	-----



Pixel Data

0	0	0	1	2	0	...	etc (list of indices)
---	---	---	---	---	---	-----	-----------------------

$pxWidth \times pxHeight$

pxWidth
pxHeight
bits Per Pixel

Indices

```

1 CRLE
2 #DEBUT#LE MÉDECIN VOLANT#COMÉDIE#CRLE
3 CRLE
4 Des personnages dont le caractère est convenu, le costume arrêtéCRLE
5 d'avance, le langage différent, le type invariable, et qui, sur un planCRLE
6 tracé, improvisent un dialogue pittoresque, conforme aux situations,CRLE
7 CRLE
8 #FIN#LE MÉDECIN VOLANT#CRLE
9 CRLE
10 CRLE
11 #DEBUT#LA JALOUSIE DU BARBOUILLÉ#COMÉDIE#CRLE
12 CRLE
13 Le jeune Poquelin sortait du collège des jésuites et des leçons deCRLE
14 Gassendi. Frais émoulu de ses classes, il riait, avec Bernier etCRLE
15 Chapelle, du _Ferio Darii Bamalipton_ et de l'inutile parlage desCRLE
16 docteurs scolastiques; il leur préférerait, au grand scandale de saCRLE
17 famille, Tabarin et Guillot Gorju.CRLE
18 CRLE
19 #FIN#LA JALOUSIE DU BARBOUILLÉ#CRLE
20 CRLE
21 CRLE

```

```

std::vector<exo5::Play> exo5::FindAllPlays(String const& molieres)
{
    std::vector<exo5::Play> plays;

    String remaining{molieres};
    while (true)
    {
        // COMMENTAIRE
        size_t idx = remaining.find("#DEBUT#");
        if (idx == String::npos)
            break;
        exo5::Play& play = plays.emplace_back();

        // COMMENTAIRE
        remaining = remaining.substr(idx + 7);

        // COMMENTAIRE
        idx = remaining.find("#");
        17
        // COMMENTAIRE
        play.name = remaining.substr(0, idx);

        // COMMENTAIRE
        remaining = remaining.substr(idx + 1);
        17 + 1

        // COMMENTAIRE
        7
        idx = remaining.find("#");
        play.kind = remaining.substr(0, idx);
        remaining = remaining.substr(idx + 1);
        7 + 1

        // COMMENTAIRE
        ~So
        idx = remaining.find("#FIN#");
        play.content = remaining.substr(0, idx);
        remaining = remaining.substr(idx + 5);

        // COMMENTAIRE
    }

    return plays;
}

```

